

Modeling Quadruped Leg Dynamics on Deformable Terrains using Data-driven Koopman Operators

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Abstract: Quadruped locomotion over soft deformable terrain is a challenging problem. It is difficult to accurately model the nonlinear dynamics associated with the terrain-leg interaction during the stance phase of the gait motion. In order to overcome this challenge, we utilize Koopman spectral theory to obtain a linear dynamical system in the space of observable functions. We propose an experimental framework to obtain a data-driven Koopman model of the quadruped's leg dynamics over deformable terrains as a switched dynamical system. Experimental results show that the learned switched system model can be used to predict gait trajectories on unknown terrain. Furthermore, we exploit the linearity of the Koopman operator to extract the complex leg-terrain interaction dynamics. Finally, we show that the Koopman generator has a unique spectrum associated with each terrain, making terrain classification possible without the need for foot force sensors.

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Keywords: Quadruped locomotion, switched systems, data driven control, Koopman theory

1. INTRODUCTION

Quadruped locomotion in natural outdoor environments is a collection of interconnected tasks consisting of perception, state estimation, dynamic modeling, and control. Ways to increase the reliability of locomotion require improvements in any of the four aforementioned areas. For the approach of higher fidelity models for locomotion, the dynamic model of the quadruped is modeled as a hybrid system with continuous dynamics of swing and stance phases, with the switch of the system instantiated by the ground reaction force. For locomotion on a flat rigid surface, the contact forces are impulsive in nature. During the stance phase, when legs come into contact with the ground, there is a sudden discontinuous change in the joint velocities and torques while the joint positions remain constant. Hence, for the next swing phase, the robot state trajectories evolve from a different initial condition arising at the end of such contacts. These dynamics can be captured using a hybrid model with continuous dynamics and discrete events describing the evolution of robot trajectories as discussed in Ames (2014) and Ma et al. (2019). For soft deformable terrains like sand, gravel, mud etc., due to the compliance effects of the ground, the contact forces have a more prolonged evolution. The joint velocities and torques tend to be continuous before and after contact. These systems can be modeled as switched dynamical systems, similar to the formulation presented in Katayama and Ohtsuka (2022), Chen et al. (2022). Switched systems are a class of hybrid systems with no discontinuities in the trajectories and powerful tools from optimal control

and switched systems theory can be leveraged to design controllers with formal stability guarantees as discussed in Bengua and DeCarlo (2005) and Zhu and Antsaklis (2015).

Although a switched system model aids in the defining a control scheme for locomotion, another key component in locomotion is the prediction of contact forces over unknown terrain. For soft terrains, these external disturbances pose a substantial challenge for robust planning and control. Using specialized sensors could address some of these issues, however these sensors can degrade over time when subjected to repeated periodic impacts, leading to considerable inaccuracies as discussed in Cong et al. (2020) and Hutter et al. (2014). State estimation methods have been employed by Sebastian et al. (2019) and Fink and Semini (2020) to address these issues. However, it was noted by Fahmi et al. (2021) and Wisth et al. (2019) that these algorithms were developed primarily for rigid surfaces and don't perform well in soft outdoor terrain.

More advanced model-based approaches were used in Zhang et al. (2021) and Bledt et al. (2018) to represent soft ground behavior as a compliant spring and damper system. However, due to the complex nature of robot dynamics, these model-based estimation methods cannot fully capture nonlinearities and the unmodeled dynamics between the robot and the environment. Neural network based approaches were used in Cong et al. (2020), Hu and Xiong (2017) to address this issue to some extent. Its universal function approximation capabilities can be leveraged to capture the complex nonlinear relationship between the robot's joint states, input torques and contact

forces. However, these approaches don't account for the hybrid nature of quadruped dynamics.

Inspired by the recent success in Koopman based system identification for nonlinear systems reported by Bruder et al. (2019), Mauroy and Goncalves (2019), we apply Koopman theory to identify a linear switched system model for quadruped locomotion on deformable terrain. In particular, we obtain a data-driven model by approximating the Koopman generators associated with the flight and stance phase. The model is identified from experimental data generated by a quadruped robot.

2. PRELIMINARIES AND NOTATIONS

Notation: $\mathbb{R}^n$ denotes the n dimensional Euclidean space and $\mathbb{R}_{\geq 0}$ is the positive orthant. Given $\mathbf{X} \subseteq \mathbb{R}^n$ and $\mathbf{Y} \subseteq \mathbb{R}^m$, let $\mathcal{L}_1(\mathbf{X}, \mathbf{Y})$, $\mathcal{L}_\infty(\mathbf{X}, \mathbf{Y})$, and $\mathcal{C}^k(\mathbf{X}, \mathbf{Y})$ denote the space of all real valued integrable functions, essentially bounded functions, and space of k times continuously differentiable functions mapping from $\mathbf{X}$ to $\mathbf{Y}$ respectively. $\mathbf{s}_t(\mathbf{x}, \mathbf{u})$ denotes the solution of dynamical system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ starting from initial condition $\mathbf{x}_0$.

2.1 Koopman Operators and Generators

Consider a dynamical system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x} \in \mathbf{X} \subseteq \mathbb{R}^n, \mathbf{u} \in \mathbf{U} \subseteq \mathbb{R}^m \quad (1)$$

where the vector field is assumed to be $\mathbf{f}(\mathbf{x}) \in \mathcal{C}^1(\mathbf{X}, \mathbb{R}^n)$. The nonlinear dynamics in the state space can be lifted to an infinite dimensional space of functions using the Koopman operator defined as follows.

The Koopman operator $\mathbb{U}_t : \mathcal{L}_\infty(\mathbf{X}) \rightarrow \mathcal{L}_\infty(\mathbf{X})$ for dynamical system (1) is defined as

$$[\mathbb{U}_t \varphi](\mathbf{x}, \mathbf{u}) = \varphi(\mathbf{s}_t(\mathbf{x}, \mathbf{u})), \quad \varphi \in \mathcal{L}_\infty. \quad (2)$$

The infinitesimal generator for the Koopman operator

$$\lim_{t \rightarrow 0} \frac{\mathbb{U}_t \varphi - \varphi}{t} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \cdot \nabla \varphi(\mathbf{x}, \mathbf{u}) =: \mathcal{K}_f \varphi. \quad (3)$$

Since $\mathbb{U}_t$ is semi-group with generator $\mathcal{K}_f$ it satisfies

$$\frac{d}{dt} \mathbb{U}_t \varphi = \mathcal{K}_f \mathbb{U}_t \varphi. \quad (4)$$

Next we describe algorithms for the finite dimensional approximation of Koopman operator.

2.2 Data-driven Approximation of Koopman Operator

Extended dynamic mode decomposition (EDMD) method developed in Kevrekidis et al. (2016) is one of the popular algorithms for the data-driven approximation of the Koopman operator. The basic idea of the algorithm can be explained as follows. For the continuous-time dynamical system (1), consider snapshots of time-series data from single or multiple trajectories

$$\mathcal{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M], \quad \mathcal{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_M], \quad (5)$$

where $\mathbf{x}_i \in \mathbf{X}$ and $\mathbf{y}_i \in \mathbf{X}$. The pair of data sets are assumed to be two consecutive snapshots i.e., $\mathbf{y}_i = \mathbf{s}_{\Delta t}(\mathbf{x}_i)$, where $\mathbf{s}_{\Delta t}$ is solution of (1) with Δt the discretization time-step. Let $\Psi = [\psi_1, \dots, \psi_N]^\top$ be the choice of dictionary functions. For DMD, we select the unit basis vectors

$\mathbf{e}_i \in \mathbb{R}^N$, such that $\Psi = [\mathbf{e}_1, \dots, \mathbf{e}_N]^\top$. The EDMD algorithm provides a finite-dimensional approximation of the Koopman operator as the solution of the following least square problem

$$\min_{\mathbf{K}} \|\mathbf{G}_1 \mathbf{K} - \mathbf{G}_2\|_F^2 \quad (6)$$

With $\mathbf{K}, \mathbf{G}_1, \mathbf{G}_2 \in \mathbb{R}^{N \times N}$, $\|\cdot\|_F$ is the Frobenius norm. The matrices are computed from data

$$\begin{aligned} \mathbf{G}_1 &= \frac{1}{M} \sum_{m=1}^M \Psi(\mathbf{x}_m) \Psi(\mathbf{x}_m)^\top \\ \mathbf{G}_2 &= \frac{1}{M} \sum_{m=1}^M \Psi(\mathbf{x}_m) \Psi(\mathbf{y}_m)^\top \end{aligned} \quad (7)$$

The solution to the least square problem is

$$\mathbb{U}_{\Delta t} \approx \mathbf{K}_{edmd} = \mathbf{G}_1^\dagger \mathbf{G}_2. \quad (8)$$

Where $\dagger$ stands for pseudo-inverse. The convergence of EDMD towards the true Koopman operator as the number of data points and basis functions go to infinity are provided in Korda and Mezić (2018), Klus et al. (2020). The EDMD-based approximation of the Koopman operator can be used to approximate the Koopman generator as follows,

$$\mathcal{K}_f \approx \frac{\mathbf{K} - \mathbf{I}}{\Delta t} =: \mathbf{L}. \quad (9)$$

where $\mathbf{I}$ is the identity matrix. Now we introduce the nonlinear dynamics of a quadruped leg.

2.3 Quadruped Leg Dynamics

The equations of motion of a system composed of rigid elements can be described using the Euler-Lagrange equations. The individual leg configuration space is defined using $\mathbf{q} = [\theta, \gamma]^\top$ where θ is the angle of the leg with respect to the vertical, and γ is the leg separation angle as shown in Figure 1. The flight phase dynamics of the quadruped can be described using the Euler Lagrange equations as follows,

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{H}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{B} \mathbf{u} + \mathbf{J}(\mathbf{q})^\top \lambda \quad (10)$$

where $\mathbf{M}(\mathbf{q})$ is the inertia matrix, $\mathbf{H}(\mathbf{q}, \dot{\mathbf{q}})$ included the Coriolis and gravity terms, $\mathbf{B}$ is the actuation matrix, $\mathbf{u}$ is the control input (joint torques), $\mathbf{J}$ is the Jacobian of holonomic constraints and λ is the corresponding foot forces that arise due to contact. This can be written in state space form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ where the state vector is given by $\mathbf{x} = [\mathbf{q}, \dot{\mathbf{q}}]^\top$

$$\frac{d}{dt} \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix} = \begin{bmatrix} \dot{\mathbf{q}} \\ \mathbf{M}^{-1}(\mathbf{q}) [\mathbf{J}(\mathbf{q})^\top \lambda - \mathbf{H}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{B} \mathbf{u}] \end{bmatrix} \quad (11)$$

For the flight phase, there is no ground contact, therefore $\lambda = 0$ and joint torques $\mathbf{u}_f$ only consists of feed forward torques designed to track a desired trajectory $\mathbf{q}_{des}(t)$ according to a high level planner. For the stance phase, when the leg is in contact with the ground, $\lambda \neq 0$ and the joint torques $\mathbf{u}_s$ has to account for the additional feedback effects due to the disturbances from the foot contact force. Next we introduce the quadruped leg kinematics.

2.4 Quadruped leg Kinematics

The quadruped robot uses PD control to attain the desired leg angle (θ) and leg separation angle (γ). The output of

the controller is the set of torques for the desired setpoints. Inverse kinematics are derived to perform real time calculations of the motor torques needed to place the foot at a given point on its trajectory (in Cartesian coordinates). Utilizing the symmetry of the five-bar linkage of the legs, the inverse kinematics can be simplified as a two-bar linkage centered about the axis of symmetry. The leg Jacobian $\mathbf{J}_l$ for the quadruped robot can be computed as follows,

$$\begin{bmatrix} \dot{X} \\ \dot{Y} \end{bmatrix} = \begin{bmatrix} -l_1 \sin \phi_1 - l_2 \sin \psi_2 - D_1 & D_1 \\ l_1 \cos \phi_1 - l_2 \cos \psi_2 - D_2 & D_2 \end{bmatrix} \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{bmatrix} = \mathbf{J}_l \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{bmatrix}$$

$$D_1 = -l_2 \frac{-1 - 2 \sin \Delta_\phi \beta}{2 \sin \Delta_\phi} \sin \psi \quad (12)$$

$$D_2 = l_2 \frac{1 + 2 \sin \Delta_\phi \beta}{2 \sin \Delta_\phi} \cos \psi \quad (13)$$

where $\Delta_\phi = \phi_2 - \phi_1$, and

$$\beta = \frac{l_1}{2y\sqrt{2(1 - \cos \Delta_\phi)(1 - \frac{l_1(1 - \cos \Delta_\phi)}{2l_2})}}. \quad (14)$$

The leg Jacobian can be used to convert the applied motor torques to the ground reaction forces on the foot using the leg Jacobian. This relationship is given by,

$$\begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} (\mathbf{J}_l^T)^{-1}. \quad (15)$$

Motor torques are not being directly measured on the quadruped, so the estimated torques are approximated by the following,

$$\tau_m = K_t I_m \quad (16)$$

where K_t is the motor torque constant, and I_m is the armature current of the motor. The kinematics for a quadruped with this leg linkage is detailed in Blackman et al. (2016).

2.5 Switched Systems model

A switched dynamical system can be modeled as a set of indexed subsystems of the form $\dot{\mathbf{x}} = \mathbf{f}_j(\mathbf{x}, \mathbf{u}_j)$. Here, $j = \{1, 2\}$ is the subsystem index and $\mathbf{u}_j$ is the corresponding control input. A switching sequence is a list consisting of time instants τ_i and corresponding indices j_i which determines a logical transition to the subsystem j_i at the time instant τ_i . A switching sequence can be modeled as $\mathbf{s} = ((\tau_0, j_0), (\tau_1, j_1), \dots, (\tau_N, j_N))$, $0 \leq N < \infty$, $t_0 \leq \tau_0 \leq \tau_1 \leq \tau_N \leq t_f$. The states are assumed to be continuous at the switching instants, i.e., $\mathbf{x}(\tau^+) = \mathbf{x}(\tau^-)$.

In the case of quadruped locomotion, the continuous dynamics of the flight and stance phase can be defined as,

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}_f(\mathbf{x}, \mathbf{u}_f) \\ \dot{\mathbf{x}} = \mathbf{f}_s(\mathbf{x}, \mathbf{u}_s) \end{cases} \quad (17)$$

where $\mathbf{f}_f$ and $\mathbf{f}_s$ define the flight and stance phase dynamics respectively. The time instant τ_i at which the height of the foot from the ground $h_z(t)$ crosses zero can be used to construct a switching sequence $\mathbf{s}$ as follows,

$$\mathbf{s}(\tau_i, j_i) = \begin{cases} (\tau, f) & | h_z(\tau^-) < 0, h_z(\tau^+) > 0 \\ (\tau, s) & | h_z(\tau^-) > 0, h_z(\tau^+) < 0 \end{cases} \quad (18)$$

where f and s are the subsystem indices of flight and stance phase respectively.

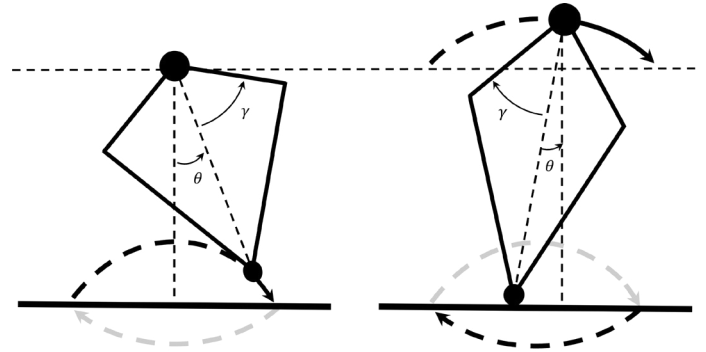


Fig. 1. Two independently driven links connected through a symmetric linkage whose generalized coordinates are chosen as the leg separation angle γ , and the angle of the leg with respect to the vertical axis θ .

3. COMPUTATIONAL METHODS

Consider the two fundamental states of contact depicted for an individual leg in Figure 1, where the left figure shows the leg in flight, and right figure shows the leg in contact with the ground. The motion associated with this leg can be fully described through the generalized coordinates θ and γ . For this robot, the control enters through the quasi-direct drive, coaxial mechanism, manifesting as two opposing torques entering through the top links. We choose to define the inputs as the motor currents I_{m0} and I_{m1} . During the stance phase, the normal force is modeled as a disturbance to the system, with F_{tot} denoting the total estimated normal force as measured through the reflected motor torques. Altogether, these measurements will be used to identify the Koopman operators associated with the flight and stance dynamics.

As most dynamical systems are naturally continuous, we will utilize the continuous time counterpart to the discrete Koopman operator, namely the Koopman generator defined in (9). We use this to represent our quadruped model and observe a subspace $\mathcal{H}$ to find $\dot{\mathbf{z}}$

$$\dot{\mathbf{z}} = \mathbf{L}_f \mathbf{z} \iff \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u} \quad (19)$$

$$\dot{\mathbf{z}} = \mathbf{L}_s \mathbf{z} \iff \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u} + \mathbf{h}(\mathbf{x})\mathbf{w} \quad (20)$$

In order to identify the dynamics associated with (19), we use EDMD. Let $\mathbf{z}$ be the observable vector such that $\mathbf{z} = [\mathbf{q} \ \dot{\mathbf{q}} \ \Psi(\mathbf{x})]^T$, where $\mathbf{q} = [\theta \ \gamma]^T \in \mathbb{R}^n$ and $\mathbf{x} = [\mathbf{q} \ \dot{\mathbf{q}}]^T \in \mathbb{R}^{2n}$. The dictionary function Ψ acts on $\mathbf{x}$, mapping $\mathbb{R}^{2n} \rightarrow \mathbb{R}^N$ where $N \gg 2n$. Therefore, we have that $\mathbf{z} \in \mathbb{R}^{2n+N}$. First, we consider the flight dynamics, where the generator equation can be written as,

$$\begin{bmatrix} \dot{\mathbf{z}} \\ * \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{2n+N \times 2n+N} & \mathbf{B}_{2n+N \times m} \\ * & * \end{bmatrix} \begin{bmatrix} \mathbf{z} \\ \mathbf{u} \end{bmatrix}$$

$$\mathbf{x} = [\mathbf{I}_{2n \times 2n} \ \mathbf{0}_{2n \times N}] \mathbf{z} \quad (21)$$

where $\mathbf{I}$ is the identity matrix and $\mathbf{u}$ is the sum of the feed-forward and feedback torques. Since we directly have access to the current measurements I_{m0} and I_{m1} from the motor controllers, we can let $\mathbf{u} = [I_{m0} \ I_{m1}]$. Now consider the stance dynamics, where now we include the disturbance $\mathbf{w} \in \mathbb{R}^p$ entering the system through $\mathbf{H}$.

$$\begin{bmatrix} \dot{\mathbf{z}} \\ * \\ * \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{2n+N \times 2n+N} & \mathbf{B}_{2n+N \times m} & \mathbf{H}_{2n+N \times p} \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} \mathbf{z} \\ \mathbf{u} \\ \mathbf{w} \end{bmatrix} \quad (22)$$

$$\mathbf{x} = \begin{bmatrix} \mathbf{I}_{2n \times 2n} & \mathbf{0}_{2n \times N} \end{bmatrix} \mathbf{z}$$

Once again, $\mathbf{u} = [I_{m0} \ I_{m1}]$; however, now in addition to the feed-forward torques, there are also PD feedback torques. The disturbance input is modeled as the foot reaction force where $\mathbf{w} = F_{tot}$ is the sum of the reflected forces felt by the motors to maintain the desired setpoint position. The solutions to equations (21) and (22) can be obtained using M snapshots generated from the system at a fixed sampling frequency, so as to construct the snapshot matrices,

$$\mathcal{X} = \begin{bmatrix} \mathbf{z}_1 & \dots & \mathbf{z}_{M-1} \\ \mathbf{u}_1 & \dots & \mathbf{u}_{M-1} \\ \mathbf{w}_1 & \dots & \mathbf{w}_{M-1} \end{bmatrix}, \quad \mathcal{Y} = \begin{bmatrix} \mathbf{z}_2 & \dots & \mathbf{z}_M \\ \mathbf{u}_2 & \dots & \mathbf{u}_M \\ \mathbf{w}_2 & \dots & \mathbf{w}_M \end{bmatrix}$$

Now the Koopman operator $\mathbf{K}_j$ and generator $\mathbf{L}_j$ associated with each subsystem j (corresponding to flight and stance dynamics) can be obtained using Equation (8) and (9) respectively, with a switching sequence as defined in Equation (18).

Remark 1. Note, that here we neglect the input and disturbance dynamics $*$, which is naturally realized when approximating the Koopman operator with control as described in Proctor et al. (2018) and Abraham and Murphey (2019). Furthermore, it is known that a bilinear controlled dynamical system exists in the space of functions, however for the purposes of prediction we can simply assume that the control enters linearly through some $\mathbf{B}$ matrix found using least squares.

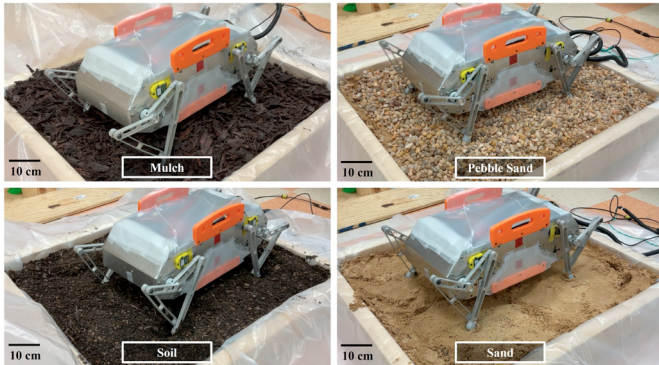


Fig. 2. The soil test bed with several terrain types used to collect data during the stance phase experiments.

3.1 Terrain Classification: Exploiting Linearity

The flight dynamics are a special case of the disturbance dynamics when $\mathbf{w} = 0$ and we are left with only the drift and control vector fields $\mathbf{f}$ and $\mathbf{g}$ respectively.

$$\dot{\mathbf{z}} = \mathbf{L}_f \mathbf{z} \approx \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u} + \mathbf{h}(\mathbf{x})\mathbf{0} \quad (23)$$

When $\mathbf{w} \neq 0$ the leg is in contact with the ground.

$$\dot{\mathbf{z}} = \mathbf{L}_s \mathbf{z} \approx \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u} + \mathbf{h}(\mathbf{x})\mathbf{w} \quad (24)$$

Its easy to see through the exploitation of linearity that we could approximate the vector field $\mathbf{h}(\mathbf{x})$.

$$\mathbf{L}_s - \mathbf{L}_f \approx \mathbf{h}(\mathbf{x}) \quad (25)$$

In the next section, we will outline the experimental procedure for gathering the measurement data.

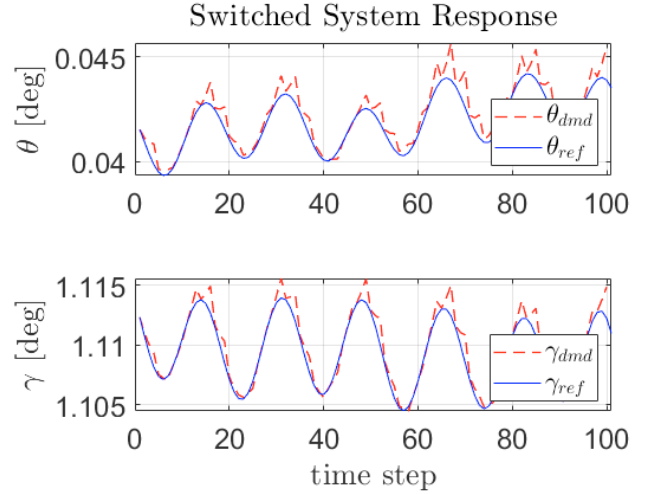


Fig. 3. Filtered DMD predicted state response for the switched dynamical system predicting $N = 5$ time steps (200ms) in the future with a time discretization of $\Delta t = 40ms$.

4. EXPERIMENTAL METHODS

To demonstrate the effectiveness of the operator based modeling of quadruped dynamics, we performed experiments with our robot off-ground and on-ground. The detail of these tests are discussed at the end of this section. First, we introduce the hardware platform.

4.1 Hardware Description

The mechanical structure of the robot used in this paper is based on the Stanford Doggo developed by Kau et al. (2019). The body of 20 cm wide and 13 cm in height with a mass of 6 kg. The legs of the robot are designed as a symmetric five-bar linkage with a length of 9 cm and 16 cm for the upper and lower links respectively.

An off-board computer was used to send commands to the Teensy 3.5, the on-board microcontroller, via USB serial communication. The joint states and control inputs were sampled at 35 Hz through real time ROS messages using an XBee wireless module. Odrive 3.5 motor controllers are used to receive leg position commands and actuate the motors accordingly. Low level control is achieved through a PD controller running at 100 Hz to execute the desired gait trajectories.

4.2 Flight Phase Experiment

To study the quadruped leg dynamics at flight phase, the robot was mounted on a test bench for the entire gait sequence. We sampled open-loop gait trajectories from the following gaits: pronk, bound, walk, turn trot, and dance. The only torque required to move the leg during this experiment was the feed forward torque $\mathbf{u}_f$ based on the flight phase dynamics modeled as $\dot{\mathbf{x}} = \mathbf{f}_f(\mathbf{x}, \mathbf{u}_f)$.

4.3 Stance Phase Experiment

For the stance phase experiments, the robot was set on different terrain such as rigid ground, sand, pebbles on

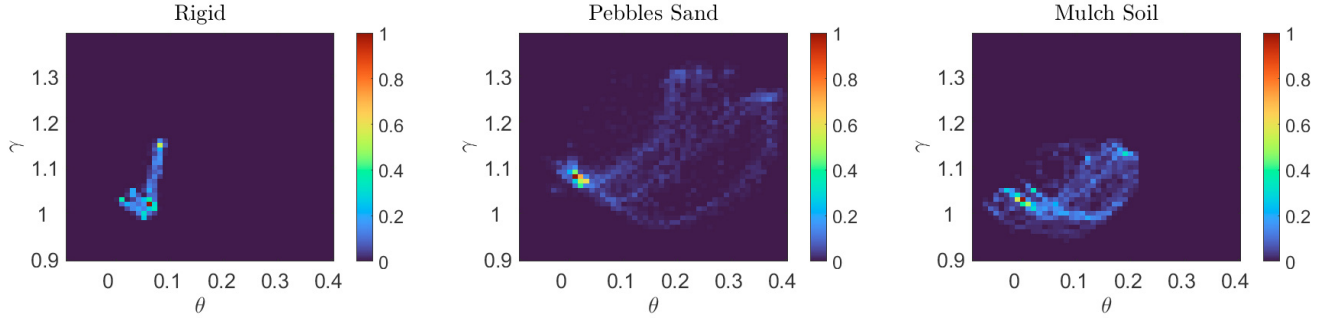


Fig. 4. The variation in terrain results in distinct occupancy maps showing the space occupied by all leg angles θ and γ in an invariant subspace.

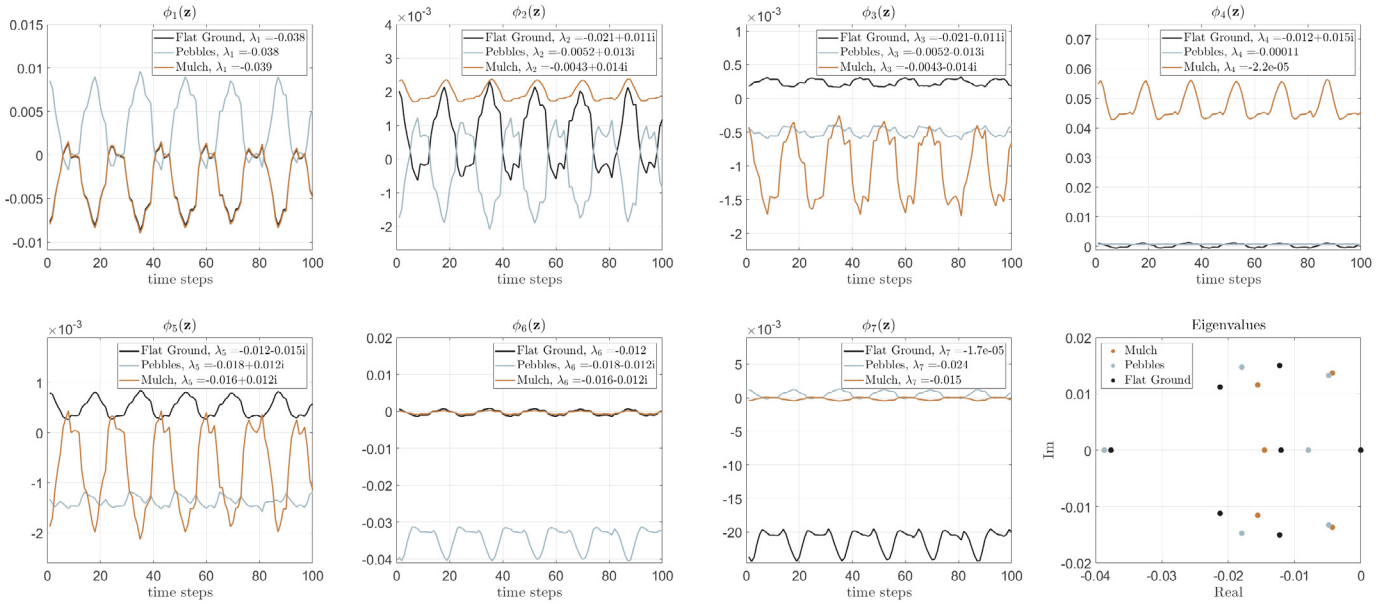


Fig. 5. Dominant eigenfunctions of a leg obtained from DMD corresponding to each terrain. (Last): Eigenvalues of the generator $\mathbf{L}_s$ associated with the each terrain type. (using a γ gain of $K_p = 30$).

sand, soil, and mulch on soil as shown in Figure 2. To identify the stance phase dynamics $\dot{\mathbf{x}} = \mathbf{f}_s(\mathbf{x}, \mathbf{u}_s, \mathbf{w})$ We execute a static "dance" gait where the robot flexes up and down with the foot in full contact with the ground. Tests were conducted using 3 different proportional gains for γ ($K_p = \{25, 30, 40\}$). This results in a range of contact force responses acting on the robot foot depending on the terrain properties. Now the torque required to move the leg up and down, $\mathbf{u}_s$, is sum of feed forward torque and the feedback PD torque.

5. RESULTS AND DISCUSSION

Experimental results with the Clemson Tigger quadruped shows that the switched system model learned through a data driven Koopman framework is able to predict unseen gait trajectory on unknown terrains. The learned system model is validated on unseen data, as shown in Figure 3, where the model is able to accurately predict the future dynamics over a finite time horizon. We can analyze the invariant subspaces as shown in Figure 4. Using the flat ground as a baseline comparison, it can be inferred that the mulch-soil and pebbles-sand terrains

cause perturbations in θ due to terrain-leg interaction dynamics. This information alone could be used to help classify the terrain type.

Soft terrains, such as mulch-soil, shift the eigenvalues towards the real axis, resulting in a greater damping in the legs response. Conversely, stiffer terrain such as pebbles-sand, has less of a damping effect. These distinctions can be seen through the eigenfunctions of the generator (see Figure 5). Furthermore, different terrain properties manifest themselves through the shift of eigenvalues in the Koopman generator shown in Figure 5. Once again, we use the eigenvalues of the flat ground as a baseline model for comparison. Each terrain excites specific eigenmodes of the learned Koopman generator, which can be utilized for terrain classification. For example, mulch-soil has larger variability associated with $\phi_3(\mathbf{z})$ and $\phi_5(\mathbf{z})$. For pebbles-sand there is a clear separation in response compared to the other terrains corresponding to $\phi_6(\mathbf{z})$. Thus, our approach provides a framework to classify terrains through the spectrum of the generator.

6. CONCLUSIONS

In summary, we apply Koopman operator theory to learn the switched system dynamics of quadruped locomotion on deformable terrains. The learned switched system model can accurately predict unseen gait patterns on unknown terrain. Although, the experimental setup requires two sets of test to distinguish the flight and stance phases, this could also be accomplished through a single training phase through segmentation of contact states gathered from walking data. As we demonstrated, the spectrum of the Koopman generator is unique for each terrain. The generators of deformable terrains exhibit eigenvalues closer to the real axis when compared to rigid ground. Furthermore, we demonstrated that different terrains excite different modes through the response of the eigenfunctions. The proposed framework relies only on the joint state and control data, with the additional benefit that the tests do not require any specialized sensors or complex state estimation methods.

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